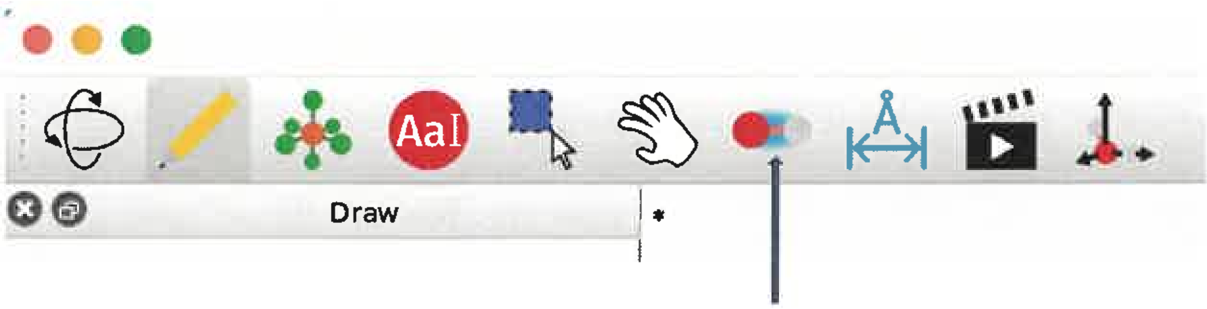
ACMP100

Quiz 1 (Marks 20)

(1) Name a forcefield in Avogadro used for the minimization of energy? (1)

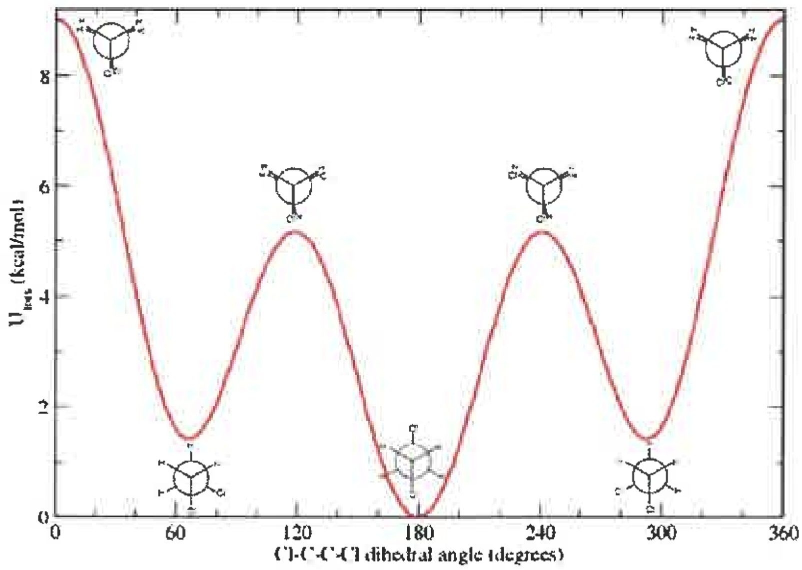
MMFF / UFF / rev-BPE

(2) Name the tool shown by arrow in Avogadro software? (1)



Bond Centric manipulation took

(3) Answer the questions based on the Figure given below for the conformation of 1,2-dichloroethane (1 x 3)



(a) Cl-C-C-Cl dihedral angles at local minima are 60-70 and 290-300

(b) The Cl-C-C-Cl dihedral angle at global minima is 180

(c) The conformation of the molecule for which the Cl-C-C-Cl dihedral angle is zero is called ____________ conformation.

Eclipsed

(4) Fill in the blank (1 x 8)

(i) Number of electrons in π_{2p} orbital of N_2 is 4

(ii) Mulliken charge for the acidic hydrogen of ethanoic acid is greater than Mulliken charge for the acidic hydrogen of propanoic acid (Smaller/greater)

(iii) The term added in the ORCA input for getting IR spectra is FREQ

(iv) The term added in the ORCA input for looking at the effect of water as solvent is CPCM(WATER)

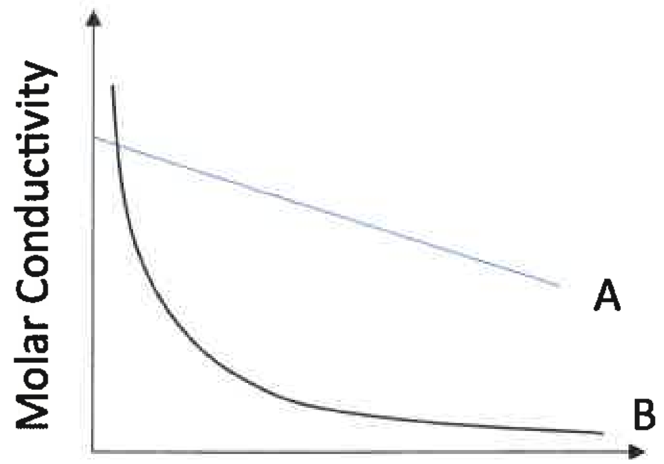
(v) Scattering of Blue light by dust particle is greater than the red light (greater/smaller)

(vi) Complementary color of green is RED

(vii) If conjugation increases in unsaturated compound, the wavelength of light increases (increases/decreases)

(viii) If the intensity of the light becomes half of the intensity after absorption from a sample, the absorbance of the sample is ~~log~~ Log 2

(5) Fill in the blank based on following figure for molar conductivity of two electrolytes A and B (1x3)



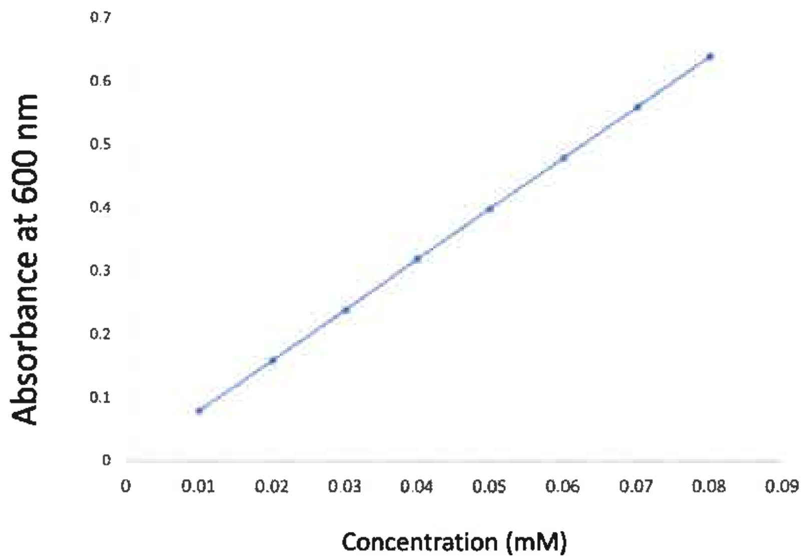
(i) Electrolyte A is Strong electrolyte. (Strong/Weak)
(ii) Electrolyte B is Weak (Strong/weak)

(iii) $\sqrt{C}$ is plotted on the x-axis.

↓

Concentration

(6) Take a look at the calibration curve (Absorbance at 600 nm Versus Concentration plot) of a dye. (1 x 2)



(i) What is the molar absorption coefficient of the dye if path length of cuvette is 1 cm (write with proper unit)? $8000\ M^{-1} cm^{-1}$

$8 \times 10^5\ M^{-1} cm^{-1}$

(ii) If absorbance of an unknown solution at 600 nm is 0.5, the concentration of the dye is 0.0625 mM

(7) Draw the molecular orbitals pictures of bonding and antibonding orbitals formed on the end to end overlap of p_x orbitals (2)